MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Infection Prophylaxis in Pediatric Oncology Guideline

*** Patient or disease state specific factors may warrant guideline deviation. Consult primary oncology team (fellow/attending/PharmD) to ensure appropriate prophylaxis is chosen. Levofloxacin, micafungin, and mold-active azoles are restricted antimicrobials that require ASP approval when deviating from this guideline. ***

I. PURPOSE: To describe pediatric oncology patients at risk for infections and outline infection prophylaxis as such patients are at risk due to myelosuppression, immunity alteration, disruption of integumentary barrier integrity, changes in colonizing microflora, and undernourishment.

II. SCOPE: This guideline outlines routine infection prophylaxis for at risk patients based upon primary oncologic diagnosis. The chart below depicts disease specific indications for prophylaxis if not otherwise stated in the patient's chemotherapy protocol.

Table 1: Antimicrobial Prophylaxis Indicated by Primary Diagnosis/Protocol

Oncologic Diagnosis		Viral[1]	Fungal[2]	Bacterial[2]	PJP[3]	IgG[4]
Leukemia	SR B-cell ALL/Lly				●	●
	HR B-cell ALL/Lly T-cell ALL/Lly		●	●	●	●
	Down Syndrome ALL Infant ALL Relapsed ALL/Lly Refractory ALL/Lly	●	●	●	●	●
	AML/MDS Relapsed/Refractory AML	●	●	●	●	
Hodgkin Lymphoma	New or Relapsed Hodgkin Lymphoma				●	
Non-Hodgkin Lymphoma	New or Relapsed Non-Hodgkin Lymphoma	●			●	
Solid Tumor	Non-CNS Solid Tumors CNS Solid Tumors				●	
	HEADSTART IV		●	●	●	

ALL=acute lymphoblastic leukemia; Lly = lymphoblastic lymphoma; AML= acute myelogenous leukemia; MDS= myelodysplastic syndrome; CNS= central nervous system

1. Viral:
    a. Consider Herpes Simplex Virus (HSV) prophylaxis for seropositive patients throughout chemotherapy
    b. Consider nirsevimab for Respiratory Syncytial Virus (RSV) prophylaxis per guidelines for age ≤19 months
2. Fungal/Bacterial:
    a. For diagnoses indicated above: Prophylaxis is indicated during chemotherapy cycles expected to cause prolonged (>7 days) severe neutropenia (ANC < 500/µL). This includes most AML and ALL chemotherapy cycles, except ALL maintenance and interim maintenance.
    b. Initiate prophylaxis when ANC < 500/µL (exception for HR B-cell or T-cell ALL/Lly induction: start fungal prophylaxis by day 10 regardless of ANC).
    c. Discontinue prophylaxis when ANC ≥ 200/µL and rising.
3. PJP: Prophylaxis indicated at start of therapy and continued through 3 months off therapy & ALC > 1000/µL.
4. IgG: Routine monthly surveillance recommended; supplementation with IVIG indicated when IgG < 400 mg/dL.

III. DOSING GUIDELINES AND PREFERRED AGENTS:

1. Viral prophylaxis:

Table 2: Recommended Viral Prophylactic Agents by Pathogen		
Viral pathogen	First Line	Comments
HSV	**Acyclovir** < 6 years: 200 mg PO BID ≥ 6 years: 400 mg PO BID	Intravenous formulation not recommended for routine prophylaxis
RSV	Nirsevimab (Beyfortus)	Refer to institutional Nirsevimab (Beyfortus) guidelines for dosing

2. Fungal prophylaxis:

Table 3: Recommended Fungal Prophylactic Agents by Diagnosis/Protocol			
Oncologic Diagnosis	Preferred Agent	Alternative Agent	Agent Rationale
HR B or T-cell ALL/Lly	**Micafungin** IV (inpatient) or **Fluconazole** PO/IV* (outpatient)	None	Yeast; drug PK and interactions
Down Syndrome ALL and Infant ALL	**Micafungin** IV	**Posaconazole** PO/IV* or **Voriconazole** PO/IV*	Yeast and mold; chemotherapy interactions
Relapsed ALL/Lly **Refractory ALL** **AML/MDS** **Relapsed/Refractory AML**	**Posaconazole** PO/IV*	**Voriconazole** PO/IV* **Isavuconazole** PO/IV* **Micafungin** IV **Alternatives listed in order of preference – see details below	Yeast and mold (including mucoromycetes**)
HEADSTART IV	**Micafungin** IV (inpatient) or **Fluconazole** PO/IV* (outpatient)	None	Yeast; drug PK and interactions
*See below for managing potential DDIs between azoles and vincristine ** See below for potential disadvantages of alternative agents if mucoromycetes coverage is recommended			

- Posaconazole and voriconazole are strong CYP3A4 inhibitors, and fluconazole and isavuconazole are moderate CYP3A4 inhibitors. These agents may be contraindicated or require dose modifications based on each patient's chemotherapy plan and/or clinical trial.
- Voriconazole and micafungin do not cover mucoromycetes. Isavuconazole is not the preferred agent for fungal prophylaxis due to reports of breakthrough infections. Micafungin is not available in an oral formulation. For patients who cannot receive posaconazole, selection of an alternative agent should take these limitations into account.
- Agents:
  - **Fluconazole** 6 mg/kg PO/IV daily (max: 400 mg/dose)
    - Contraindicated within 24 hours of vincristine administration due to increased neurotoxicity. Recommend holding the day prior, the day of, and the day after administration of vincristine.
  - **Micafungin** 5 mg/kg IV daily (max: 100 mg/dose)
  - **Voriconazole**
    - < 18 years: 9 mg/kg PO/IV BID (initial maximum dose: 300 mg/dose)
    - ≥ 18 years: 200 mg PO/IV BID
    - Obtain trough level after 5-7 days
      - Oral formulation preferred
      - Contraindicated within 24 hours of vincristine administration due to increased neurotoxicity. Recommend holding the day prior, the day of, and the day after administration of vincristine. For inpatients, consider substituting micafungin.

- **Posaconazole**
  - Delayed-release tablet:
    - < 10 kg: 50 mg PO BID
    - 10 to < 20 kg: 100 mg PO BID
    - 20 to < 35 kg: 200 mg PO BID
    - ≥ 35 kg: 300 mg PO BID
    - Delayed-release tablets may be crushed and diluted in water for administration by mouth or through an enteral tube (may be decreased absorption relative to oral administration). Posaconazole therapeutic drug monitoring is required when using crushed tablets.
  - Immediate-release suspension is not recommended due to frequency of dosing required (QID) and erratic/pH-dependent absorption. Delayed-release suspension currently not available due to logistical difficulties. Suspensions are NOT interchangeable for crushing delayed-release tablets.
  - Intravenous: 6 mg/kg IV BID for 2 doses followed by 6 mg/kg IV daily (max: 300 mg/dose)
  - Obtain trough level after 5-7 days
- **Isavuconazole**
  - Capsules (available in 74.5 mg and 186 mg strengths):
    - Age ≥ 6 years:
      - 16 to < 18 kg: 149 mg PO q8h x 6 doses followed by 149 mg PO daily
      - 18 to < 25 kg: 223.5 mg PO q8h x 6 doses followed by 223.5 mg PO daily
      - 25 to < 32 kg: 298 mg PO q8h x 6 doses followed by 298 mg PO daily
      - ≥ 32 kg: 372 mg PO q8h x 6 doses followed by 372 mg PO daily
    - Capsules may be opened and mixed with water, saline, or soft foods/formula for administration by mouth or through an enteral tube. Isavuconazole therapeutic drug monitoring is required when using opened capsules.
  - Intravenous:
    - Age < 3 years: 15 mg/kg IV q8h x 6 doses followed by 15 mg/kg IV daily
    - Age ≥ 3 years: 10 mg/kg IV q8h x 6 doses followed by 10 mg/kg IV daily (max: 372 mg/dose)
  - Obtain trough level after 5-7 days

3. **Bacterial prophylaxis:**

- Agent:
  - **Levofloxacin**
    - < 5 years: 10 mg/kg PO/IV BID (max: 375 mg/dose)
    - ≥ 5 years: 10 mg/kg PO/IV daily (max: 500 mg/dose)
- Anti-bacterial prophylaxis is not indicated during episodes of febrile neutropenia if not originally on prophylaxis based on Table 1

4. **PJP prophylaxis:**

Table 4: Recommended PJP Prophylactic Agents & Dosing by Age

Age Group	First Line	Second Line	Third Line
< 1 month	**Atovaquone** 30 mg/kg PO daily	**Dapsone** 2 mg/kg PO daily (max: 100 mg/dose) or 4 mg/kg PO once weekly (max: 200 mg/dose)	**Pentamidine** 4 mg/kg IV q4 weeks (max: 300 mg/dose)
≥ 1 - < 24 months	**Bactrim** 2.5 mg TMP/kg/dose PO BID 2 days per week (Sat/Sun) (max: 160 mg TMP/dose)	**Atovaquone** PO once daily <3 months: 30 mg/kg ≥3 months-<24 months: 45 mg/kg	**Pentamidine** 4 mg/kg IV q4 weeks (max: 300 mg/dose)
≥ 24 months – < 5 years		**Pentamidine** 4 mg/kg IV q4 weeks (max: 300 mg/dose)	**Atovaquone** 30 mg/kg PO daily (max: 1500 mg/dose)
≥ 5 years		**Pentamidine** 300 mg inhaled q4 weeks	**Pentamidine** 4 mg/kg IV q4 weeks (max: 300 mg/dose)

- Agents:
  - **Trimethoprim-sulfamethoxazole (Bactrim)**
    - Contraindicated within 24 hours of High Dose (HD) Methotrexate infusion, allergy to sulfa medications, known G6PD deficiency, or infants < 1 month with hyperbilirubinemia
      - Strong recommendation to adjust dosing schedule to avoid administration within 24 hours of HD-methotrexate; should not be restarted until clearance achieved per protocol
    - In the absence of a contraindication, Bactrim is strongly preferred over alternative agents due to increased incidence of breakthrough infections with alternative agents. In patients receiving HD-methotrexate, Bactrim is still preferred if dosing schedule can be adjusted as described above
  - **Pentamidine** IV
    - Due to increased breakthrough infections in patients <24 months, use of atovaquone is preferred if unable to take Bactrim
  - **Pentamidine** inhaled (patients ≥ 5 years old)
    - Must be able to do a PFT test to properly inhale pentamidine
    - Inhaled pentamidine is not recommended in patients that have received lung field radiation or have a history of lung fibrosis, or abnormal PFT
  - **Atovaquone**
    - Recommend prompt transitioning to alternative agent when safe due to poor palatability and expense
  - **Dapsone**
    - Risk of methemoglobinemia 20-30%. Contraindicated in G6PD deficiency (testing recommended)

5. **Immunoglobulin Supplementation:**
   - Agent:
     - IVIG 0.4 g/kg/dose IV

Additional Approvals:
CPC: 04/2022

Last revised: 04/2026	**Antimicrobial Stewardship Subcommittee:** 04/2026	**P&T:**

References

	Citation
1	Lehrnbecher T, Fisher BT, Phillips B, et al. Guideline for antibacterial prophylaxis administration in pediatric cancer and hematopoietic stem cell transplantation. *Clinical Infectious Diseases* 2020; 71 (1): 226-36
2	Lehrnbecher T, Fisher BT, Phillips B, et al. Clinical practice guideline for systemic antifungal prophylaxis in pediatric patients with cancer and hematopoietic stem-cell transplantation recipients. *Journal of Clinical Oncology* 2020; [ePub May 27, 2020]
3	Proudfoot R, Phillips B, Wilne S. Guidelines for the prophylaxis of *Pneumocystis jirovecii* Pneumonia (PJP) in children with solid tumors. *Journal of Pediatric Hematology Oncology* 2017; 39 :194-202
4	Maertens J, Cesaro S, Maschmeyer G, et al. ECIL guidelines for preventing *Pneumocystis jirovecii* pneumonia in patients with haematologic malignancies and stem cell recipients. *Journal of Antimicrobial Chemotherapy* 2016; 71: 2397-2404
5	Fisher B, Robsinson P, Lehrnbecher T, et al. Risk Factors for Invasive Fungal Disease in Pediatric Cancer and Hematopoietic Stem Cell Transplantation: a Systematic Review. *Journal of Pediatric Infectious Diseases Society*. 2018;7(3):191-8
6	Dutaa A, Ikwuezunma A, Castellanos M, et al. An evidence-based, risk-adapted algorithm for antifungal prophylaxis reduces risk for invasive mold infections in children with hematologic malignancies. *Pediatric Blood and Cancer*. 2021; e29228.

7	Groll A, Pana D, Lanternier F, et al. 8th European conference for infections in Leukemia: 2020 guidelines for the diagnosis, prevention, and treatment of invasive fungal diseases in paediatric patients with cancer or post-haematopoietic cell transplantation. *Lancet Oncology.* 2021; 22: 3254-69
8	Wolf J, Tang L, Flynn PM, et al. Levofloxacin prophylaxis during induction therapy for pediatric acute lymphoblastic leukemia. *Clinical Infectious Diseases.* 2017;65(11):1790-8
9	Sulis ML, Blonquist TM, Stevenson KE, et al. Effectiveness of antibacterial prophylaxis during induction chemotherapy in children with acute lymphoblastic leukemia. *Pediatric Blood and Cancer*;65:e26952
10	Ishida K, Haraguchi M, Kimura M, et al. Incidence of breakthrough fungal infections in patients with isavuconazole prophylaxis: A systematic review and meta-analysis. *Open Forum Infectious Diseases. 2025;12(4):ofaf163*

Revision History	
Date	**Revision**
04/2026	Changed indications for prophylaxis; clarified when to use bacterial/fungal prophylaxis; replaced palivizumab with nirsevimab; expanded weight-based dosing for posaconazole and isavuconazole

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.